INCH-POUND
MIL-PRF-62259B
26 June 1998
SUPERSEDING
MIL-R-62259A(AT)
28 September 1980
(See 6.3)

PERFORMANCE SPECIFICATION

RADIATOR: ENGINE COOLANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers engine coolant radiators used on Howitzer, Medium, Self-propelled, models M109 and M109A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are referenced in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in Section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may improve this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter

ASMC N/A FSC 2930

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SPECIFICATIONS

FEDERAL

A-A-870	- Antifreeze.
A-A-52624	- Antifreeze, Multi Engine Type.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

Bulletin 210	- Method of Testing Fans (DoD Adopted).

(Application for copies should be addressed to the Air Movement and Control Association, 30 W. University Drive, Arlington Heights, IL 60004.)

GENERAL MOTORS CORPORATION

GM 9540P	- Accelerated Corrosion Test.

(Application for copies should be addressed to General Motors Corporation, c/o Global Engineering Documents, 15 Inverness Way East, Englewood, CO 80112.)

2.4 Order of precedence. If there is a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.4), a sample shall be subjected to first article inspection.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified herein, the design, materials and manufacturing process selection shall be the prerogative of the contractor provided the radiator is able to serve as a drop-in replacement on howitzer models M109 and M109A1.

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3.2.1 Recycled, recovered, or environment preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided the material meets or exceeds all specified requirements and promotes economically advantageous life cycle costs.

3.3 Operating requirements.

3.3.1 Heat transfer capacity. The heat transfer capacity, based on the unobstructed area of the core, shall not be less than shown on figure 1.

3.3.2 Air flow resistance. The maximum static air flow resistance shall not be more than that shown on figure 1.

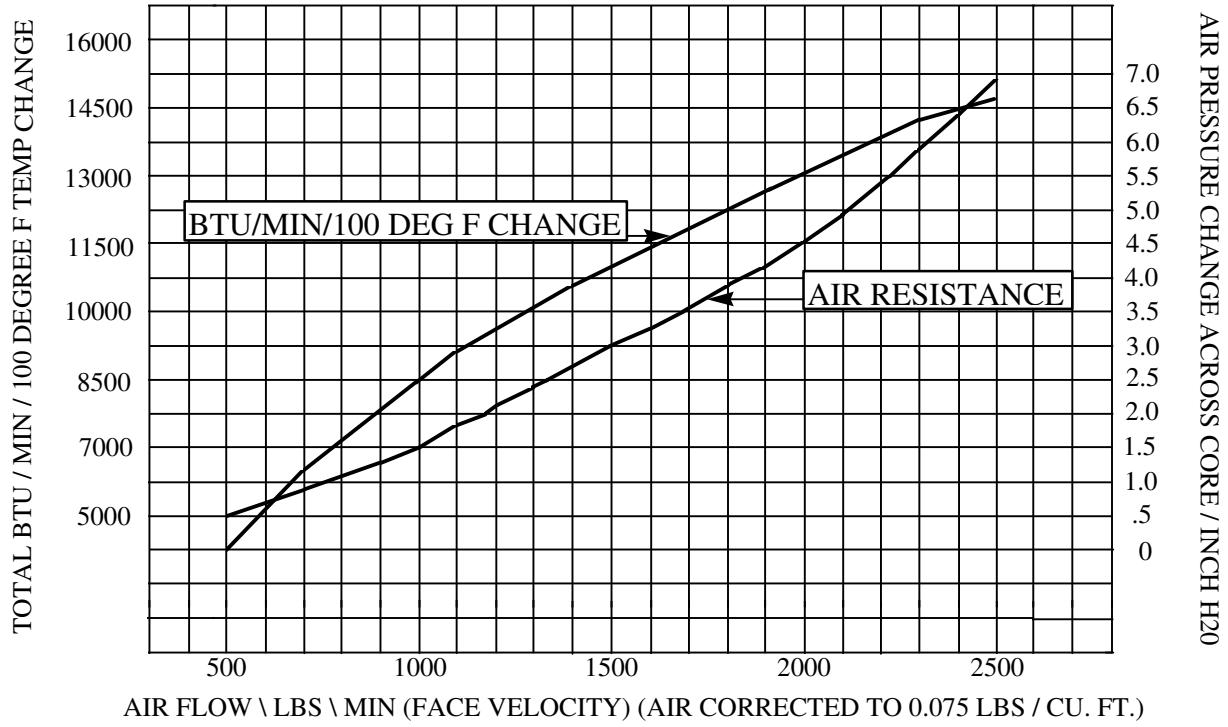


FIGURE 1. Btu rating of radiator.

3.3.3 Coolant flow resistance. The maximum coolant flow resistance at 94 gallons per minute (gal/min) (5.9 liters per second (L/s)) coolant flow rate shall not be more than 3.0 pounds per square inch (psi) (20.7 kilopascal (kPa)).

3.3.4 Pressure cycle. Radiators shall withstand a minimum of 10 000 air pressure cycles at 6 cycles per minute without rupturing or leaking.

3.3.5 Pressure capacity. Radiators shall withstand an internal pressure of 18 pounds per square in gauge (psig) (124.1 kPa) for a period of not less than 0.5 minute (or approved alternate

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regime, see 4.3.1) without evidence of rupture, permanent deformation of internal or external parts, or leakage.

3.3.6 High pressure capacity. Radiators shall withstand 25 psig (172 kPa) for 5 minutes, with not more than 3 cubic centimeters (cm^3) per minute leakage evident, when the radiator is submerged in water under test pressure. Deformation, but not rupture, is permitted.

3.4 Interface requirements.

3.4.1 Functional fit. The radiator shall serve as a drop-in replacement for the models M109 and M109A1. Complete interface and interoperability shall be achieved without engine compartment modification. The radiator shall interface and be interoperable without engine compartment modification. Interface dimensions are as indicated in figure 2 (see 6.6). Radiator mounting facilitators, bushings or brackets if used, shall be the prerogative of the manufacturer as long as all the requirements of the specification are met (see 6.6).

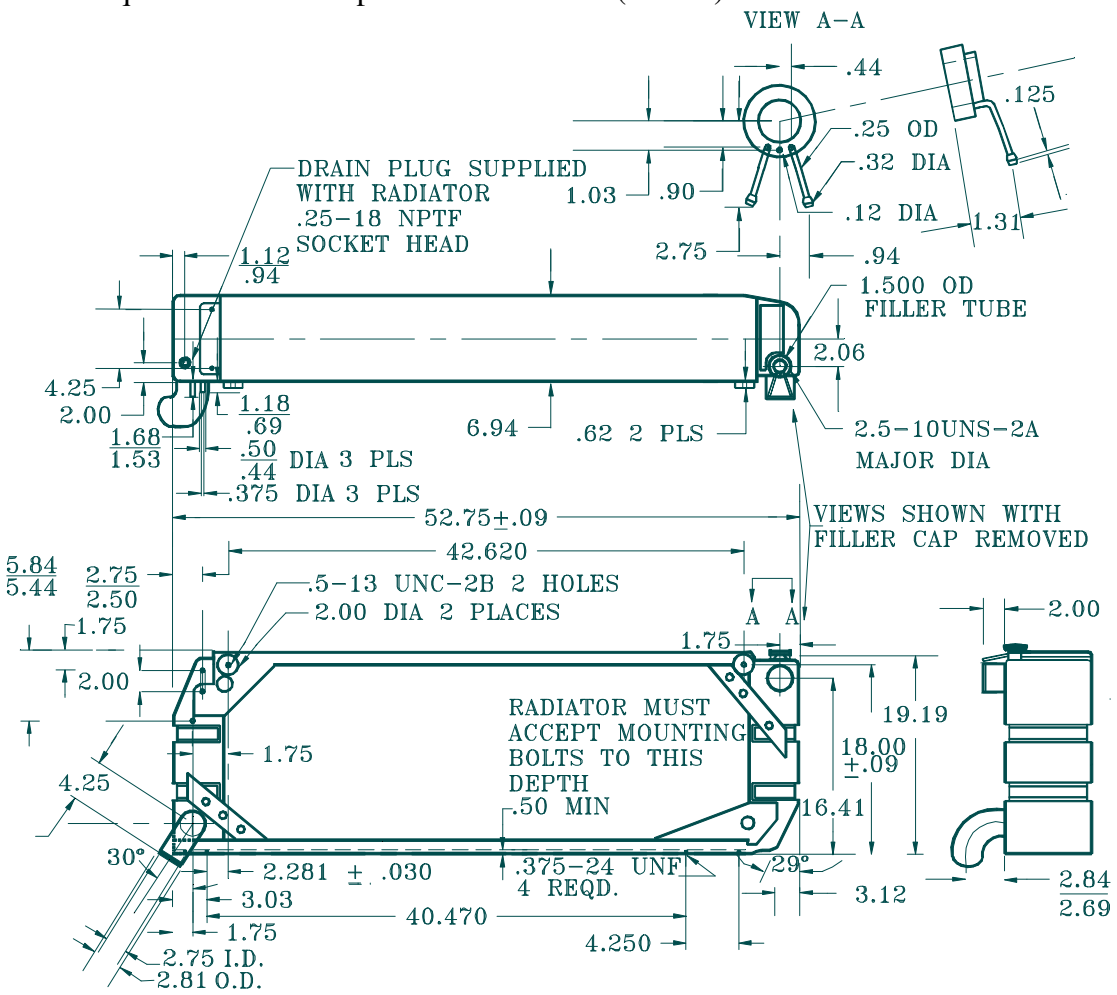


FIGURE 2. Interface and interoperability dimensions.

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3.4.2 Assembly. The manufacturer's method of assembling radiator components shall not interfere with the functional interface of the radiator pressure cap or of vehicle components necessary for cooling system operation.

3.4.3 Dissimilar metals. The radiator shall be constructed of materials designed to minimize galvanic corrosion in dissimilar metal couples.

3.5 Support and ownership requirements.

3.5.1 Safety. The radiator construction shall minimize the risks to personnel from burrs, sharp edges and avoidable scalding risks.

3.5.2 Lifting eye. The lifting eye or equivalent lifting facilitator, as installed, shall withstand a 300 pound (lb) (1335 newtons (N)) pull in any direction within 45 degrees of vertical, with the radiator in its installed position, without damage to the lifting eye or the radiator.

3.5.3 Protective finish. Radiator shall be cleaned, treated and finished per manufacturer's standard practice. The color shall be dull black and shall provide a non-brittle adhesion. Finish shall not be applied in locations where radiator performance would be inhibited.

3.5.4 Coolant level determination. The radiator shall be constructed in such a manner that with the filler cap removed, the coolant level in the radiator may be visually determined regardless of the level within the radiator.

3.5.5 Identification marking. Identification marking shall consist of the following information in permanent, easily visible, straight simple letters and numbers applied near the coolant inlet:

a. Army part number
b. National stock number
c. Manufacturer's name
d. Manufacturer's part number
e. Symbol "US"

3.5.6 Cleaning. Radiators shall be cleaned and free of contamination such as dirt, sand, metal chips, rust, cracks, foreign materials, or other defects that could impair their serviceability.

3.6 Operating environment requirements.

3.6.1 Corrosion resistance. The radiator shall be of such construction to prevent corrosion induced weakening of the metal or joints throughout its normal lifespan.

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3.6.2 Coolant. The radiator shall be compatible with antifreeze solution conforming to A-A-870 or A-A-52624, as conditions require.

3.6.3 Temperature range. The radiator, with appropriate coolant installed, shall function throughout a range of minus (-) 65 degrees Fahrenheit (°F) (-54 degrees Celsius (°C)) to plus (+) 240°F (116°C) without leakage. Subsequent to storage at -80°F (-62°C), the radiator shall function throughout the foregoing temperature range without leakage.

3.6.4 Vibration. The radiator shall withstand the vibration profile resulting from vehicle operation and howitzer recoil, without physical deformation or degradation in performance.

3.6.5 Core fin strength. Radiator core fins shall withstand the resultant forces of vehicle operation and maintenance without deformation that decreases performance.

4. VERIFICATION

4.1. Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. First article inspection shall be performed on the initial production ready or production-representative unit of an order, as required by the contract. When a first article is required (see 6.2 and 6.4), it includes all verifications listed in table I.

TABLE I. Verification methods.

Title	Requirements	Verification
Operating requirements	3.3	4.3.3
Heat transfer capacity	3.3.1	4.3.3.1
Air flow resistance	3.3.2	4.3.3.2
Coolant flow resistance	3.3.3	4.3.3.3
Pressure cycle	3.3.4	4.3.3.4
Pressure capacity	3.3.5	4.3.3.5
High pressure capacity	3.3.6	4.3.3.6
Interface requirements	3.4	4.3.4
Functional fit	3.4.1	4.3.4.1
Assembly	3.4.2	4.3.4.2
Dissimilar metals	3.4.3	4.3.4.3
Support and ownership requirements	3.5	4.3.5
Safety	3.5.1	4.3.5.1
Lifting eye	3.5.2	4.3.5.2

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TABLE I. Verification methods - Continued.

Title	Requirements	Verification
Protective finish	3.5.3	4.3.5.3
Coolant level determination	3.5.4	4.3.5.4
Identification marking	3.5.5	4.3.5.5
Cleaning	3.5.6	4.3.5.6
Operating environment requirements	3.6	4.3.6
Corrosion resistance	3.6.1	4.3.6.1
Coolant	3.6.2	4.3.6.2
Temperature range	3.6.3	4.3.6.3
Vibration	3.6.4	4.3.6.4
Core fin strength	3.6.5	4.3.6.5

4.1.2 Conformance inspection. Conformance inspection shall include the verifications listed in table I as defined by the contract (see 6.2 and 6.5). Sampling for inspection shall be as stated in the contract (see 6.2).

4.2 Order of inspection. Perform the cleaning test first, followed by the corrosion test, the remaining operating environment requirements, followed by the remaining verifications in any sequence. NOTE: Any destructive test shall be conducted last in sequence (see 4.3.6.4 g).

4.3 Verification methods. Acceptable verification methods included in this section are visual inspection, measurement, sample tests, full scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis and similarity to previously approved or previously qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques or equipment, including the application of statistical process control, tool control or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.3.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: 77°F ± 18°F (25°C ± 10°C)
b. Relative humidity: uncontrolled room ambient
c. Atmospheric pressure: site pressure

4.3.2.1 Test equipment. For the applicable verification tests, the radiator shall be connected to a test setup similar to that shown in figure 3 or figure 4. The radiator shall be supported on its normal points of support and shall not be supported on a cradle or bracket which in any way restrains the possible distortion of the radiator or core when under pressure.

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4.3.3 Operating requirements verification.

4.3.3.1 Heat transfer capacity test. To determine heat transfer capacity, the radiator shall be connected to a test setup similar to that shown in figure 3 or figure 4. The air temperature transducers shall not be more than 2 inches (in.) ((50.8 millimeter (mm)) from the core faces. Inlet and outlet coolant temperatures shall be measured within six inches (152.4 mm) of the ports. Air and coolant flow shall be measured at 25, 50, 75 and 100 percent of specified maximum air flow. Tap water shall be used as the coolant and the flow into the core shall be free of entrapped gases. Inlet coolant temperature shall be 180°F ± 5°F (82°C ± 3°C). The heat transfer capacity shall be based on the difference between the inlet coolant temperature and the inlet air temperature. To determine conformance to 3.3.1, the heat transfer capacity shall be measured on the air side in terms of British thermal units (Btu's) (kilojoule (kJ)) per minute, at 80°F (27°C) and 70 gal/min (4.4 L/s) water flow. Heat transfer values obtained from testing which are less than those shown on figure 1 shall constitute failure of the test.

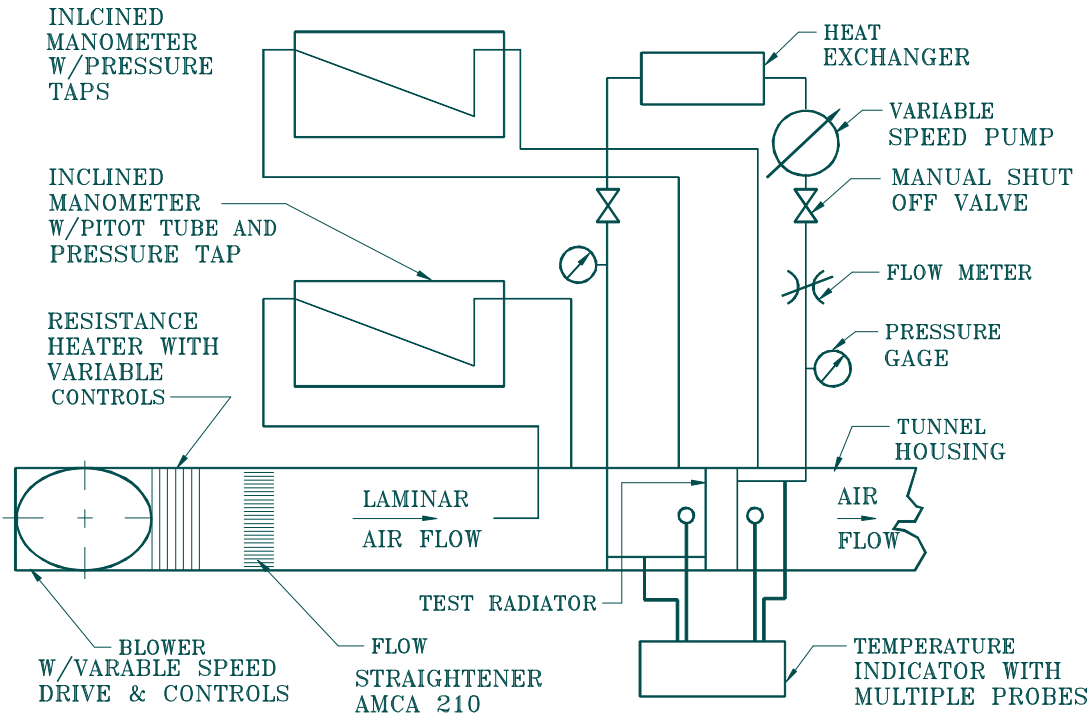


FIGURE 3. Test setup.

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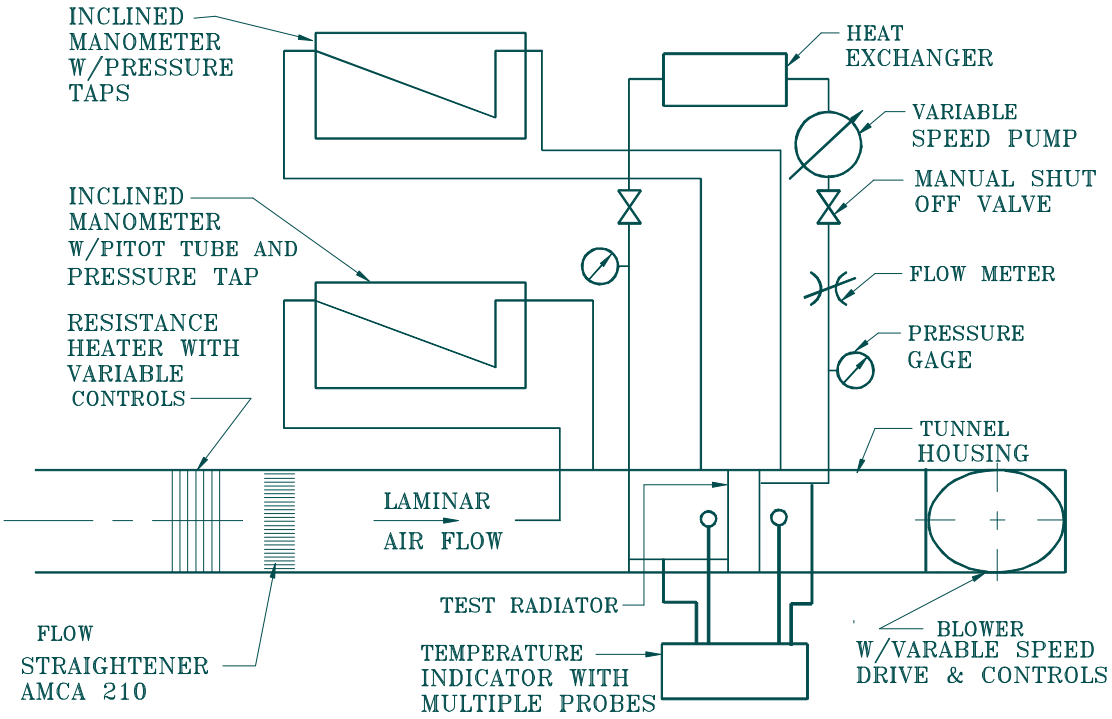


FIGURE 4. Alternate test setup.

4.3.3.2 Air flow resistance test. To determine conformance to 3.3.2, the radiator shall be connected to a test setup similar to that shown in figure 3 or figure 4. With the system operating as specified in 4.3.3.1, the air pressure drop across the radiator shall be measured at the same time as the corresponding heat capacity. The pressure taps shall be in planes parallel to, and not more than 4 in. (101.6 mm) from, the faces of the core. The air shall be measured in inches of water (kPa) at an air density of 0.075 pounds per cubic foot (lb/ft^3) (1.20 kilogram per cubic meter (kg/m^3)). The air flow resistance shall be based on the unobstructed area of the core and the average inlet velocity is to be measured within three percent. Air flow values obtained from testing which are greater than those shown on figure 1 shall constitute failure of the test.

4.3.3.3 Coolant flow resistance. To determine conformance to 3.3.3, the radiator shall be connected to a test setup similar to that shown in figure 3 or figure 4. With the system operating as specified in 4.3.3.1 and with a volume flowrate of the coolant of 94 gal/min (5.9 L/s), as measured between the radiator inlet and outlet. A measured pressure drop across the inlet and outlet ports of more than 3.0 psi (20.7 kPa) shall be cause for rejection.

4.3.3.4 Pressure cycle test. To determine conformance to 3.3.4, the radiator shall be positioned and operated in its normal vehicle position in an ethylene glycol circulatory system. Ethylene glycol shall conform to A-A-52624. The system shall maintain the ethylene glycol at a maximum temperature of 250°F ± 10°F (121°C ± 6°C) measured at radiator inlet and outlet. The radiator shall be subjected to the air pressure cycle approximated in figure 5. A cycle shall begin

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at 0 psig and rise to 18 psig (124.1 kPa), swell a minimum of two seconds above 17 psig (117.2 kPa) and return to 0 psig. At the end of the 10 000 cycles, there shall be no evidence of rupture or leakage when the radiator is pressurized to 18 psig (124.1 kPa) under water. Evidence of leakage shall be cause for rejection.

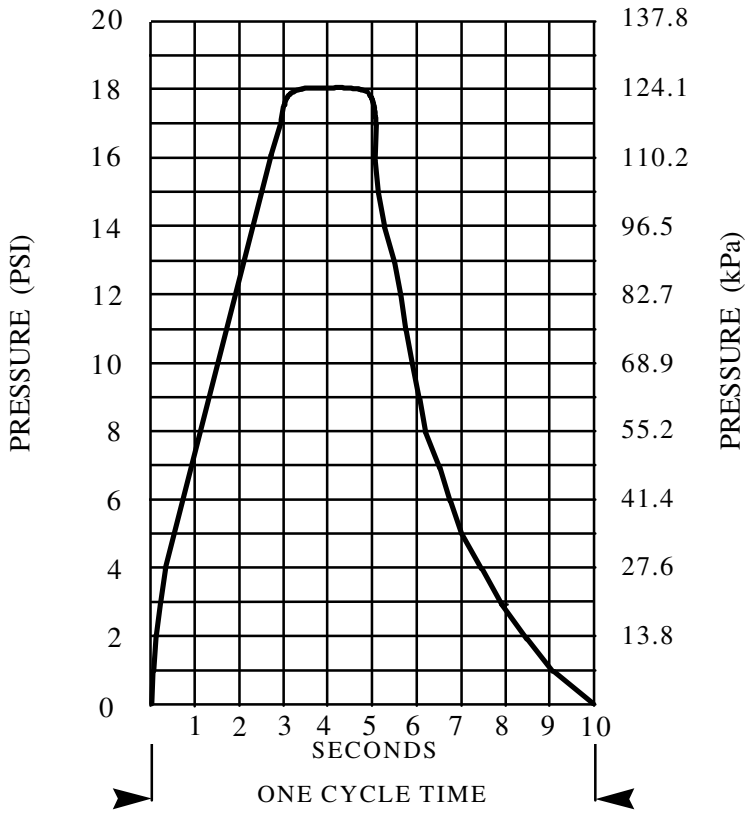


FIGURE 5. Pressure cycle.

4.3.3.5 Pressure capacity test. To determine conformance to 3.3.5, all radiators shall be subjected to a pressure capacity test. The inlet and outlet ports shall be plugged and a suitable pressure cap substituted to withstand the test pressure. An air injection nozzle shall be installed in either the inlet or outlet plug. Air pressure of 18 psig (124.1 kPa) shall be injected into the radiator and maintained for 30 seconds with radiator submerged in water. All radiators showing any sign of air bubbles by leakage, or restriction in coolant flow shall be cause for rejection. Repairs are permitted. Any radiator requiring repair shall be retested with the rejection criteria applicable as stated above.

4.3.3.6 High pressure capacity test. To determine conformance to 3.3.6, radiators shall be subjected to a high pressure capacity test. The inlet and outlet ports shall be plugged and a suitable pressure cap substituted to withstand the test pressure. An air injection nozzle shall be

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installed in either the inlet or outlet plug. Air shall be injected into the radiator to a pressure of 25 psig (172 kPa) and maintained for five minutes with radiator submerged in water. Suitable means to accurately measure escaped air shall be provided. Leakage exceeding 3 cc per minute shall be cause for rejection.

4.3.4 Interface requirements verification.

4.3.4.1 Functional fit. One radiator shall be installed in a vehicle to demonstrate that it is dimensionally compatible, both as installed and when the engine and cooling fan are operating to determine conformance to 3.4.1.

4.3.4.2 Assembly. Verification of the requirements of 3.4.2 shall consist of visual inspection, the use of gages or an applicable method from 4.3 or 4.3.1.

4.3.4.3 Dissimilar metals. The use of dissimilar metals shall be contingent on the use of suitable protection between them. Verification shall consist of visual inspection or an applicable method from paragraph 4.3 or 4.3.1.

4.3.5 Support and ownership requirements verification.

4.3.5.1 Safety. To determine conformance to 3.5.1, radiators shall be inspected for hazardous burrs, nicks, sharp edges, foreign materials or other imperfections that pose physical danger to an installer or maintenance person. Verification shall consist of visual inspection or an applicable method from paragraphs 4.3 or 4.3.1. Detection of imperfections or other risks to physical danger shall constitute failure.

4.3.5.2 Lifting eye. To determine conformance to 3.5.2, the lifting eye shall be tested by anchoring the radiator on a suitable holding fixture and applying a static load of 300 lbs (1335 N) on one eye at a time in the normally vertical axis in four directions. (Fore and aft, toward and away from the centerline of the radiator.) The load shall be applied by means of a cable, chain or rope pulling through a tension recording instrument accurate to within 5 percent and applying the load to the radiator lifting eye by means of a hook. The radiator shall then be visually examined and any failure shown by crack or deformation of the lifting eye or tank of radiator, shall be cause for rejection (see 6.7).

4.3.5.3 Protective finish. Conformance to 3.5.3 shall be determined by visual examination, use of a gloss meter, impact testing, or an applicable method from 4.3 or 4.3.1.

4.3.5.4 Coolant level determination. Conformance to 3.5.4 shall be determined by visual examination of various levels of the total radiator volume capacity, through the radiator filler neck with the cap removed. Failure of the visual examination to detect any of these coolant levels shall be cause for rejection.

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4.3.5.5 Identification marking. Conformance to 3.5.5 shall be determined by visual inspection.

4.3.5.6 Cleaning test. To determine conformance to 3.5.6, the radiator shall be visually inspected for defects. The radiator shall then be filled with distilled water or tap water of known pH index, at a temperature of 80°F to 100°F (27°C to 38°C). The radiator shall be shaken and allowed to stand for 1 hour. At the end of the 1 hour period, one-half gallon of the contents shall then be drained into a clean container and tested for pH change and chloride content. Water from the cleaned radiator showing a pH change greater than 1 shall be cause for rejection. Chlorine content which has increased more than 75 parts per million, when compared with water before rinsing, shall be cause for rejection.

4.3.6 Operating environment requirements verification.

4.3.6.1 Corrosion resistance. Compliance with 3.6.1 shall be determined by subjecting the radiator to the cyclic salt spray test, the temperature test, and the humidity test as specified in GM 9540P, Method A, Duration D. Failure of the radiator to endure the subsequent operating environment requirements herein shall be grounds for rejection.

4.3.6.2 Coolant. Compliance with 3.6.2 shall be determined by an applicable method from paragraphs 4.3 or 4.3.1.

4.3.6.3 Temperature range. To determine conformance to 3.6.3, the radiator shall be connected to a coolant flow system, the system shall flow coolant at 12 gal/min (0.79 L/s) from -65°F to 240°F (-54°C to 116°C). Radiator containing the coolant shall initially be cold soaked for 24 hours at not less than -65°F (-54°C). With the coolant flowing, the temperature shall be increased at an average rate of 1.5°F (1°C) per minute until 240°F (116°C) is reached. Additionally, subsequent to storage at -80°F (-62°C), the radiator shall function throughout the foregoing temperature range without leakage. Any sign of leakage shall be cause for rejection.

4.3.6.4 Vibration. To determine conformance to 3.6.4 the vibration test procedure shall be conducted as follows:

a. Test time at temperature of 80°F ± 10°F (27°C ± 5°C) shall be a total of 9 hours with 3 hours in each of the three mutually perpendicular axes of the radiator.
b. Radiators shall be mounted to the test fixture in accordance with the manufacturers mounting techniques.. The radiator shall be filled with water and oriented as it is installed in the vehicle.

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c. The input control sensing device(s) shall be rigidly attached to the vibration platform as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located on the test radiator to determine resonant frequencies and amplification factors. Locations to be selected should include the main structure and large component parts. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal. For sinusoidal vibration, when necessary for obtaining uniform results, a tracking filter should be used in the vibration exciter control feedback loop prior to the servo input.
d. A search shall be conducted to determine resonant frequencies along each major axis. These frequencies shall be determined by varying the frequency of applied vibration slowly through the frequency range specified on figure 6 at reduced test levels but with sufficient amplitude to excite the radiator.
e. The radiator shall be vibrated along each axis at the determined resonant frequencies. If more than three resonances are found in any one axis, the three most severe are to be chosen. Vibration shall be performed for 30 minutes in accordance with the test level specified in figure 6. Remaining time is to be utilized in the vibration sweeps test.
f. The vibration sweeps test shall be in accordance with the test level specified in figure 6. The time schedule shall be 15 minutes for sweep frequency range of 5 to 500 to 5 Hertz (Hz). When radiator resonances below 5 Hz are measured, the test curves shall be extended to 2 to 500 to 2 Hz and the sweep time shall be 18 minutes. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 7. The specified sweep time is that of an ascending plus a descending sweep time and is twice the ascending sweep time shown on figure 7 for the specified range.
g. Upon completion of vibration testing, the radiator shall be subjected to the pressure capacity test specified in 4.3.3.5. Following the pressure capacity test, sufficient physical sectioning shall be made to confirm that no failure of core bond, seam joints, or deaeration baffle assembly occur and there are no fractures of component parts.

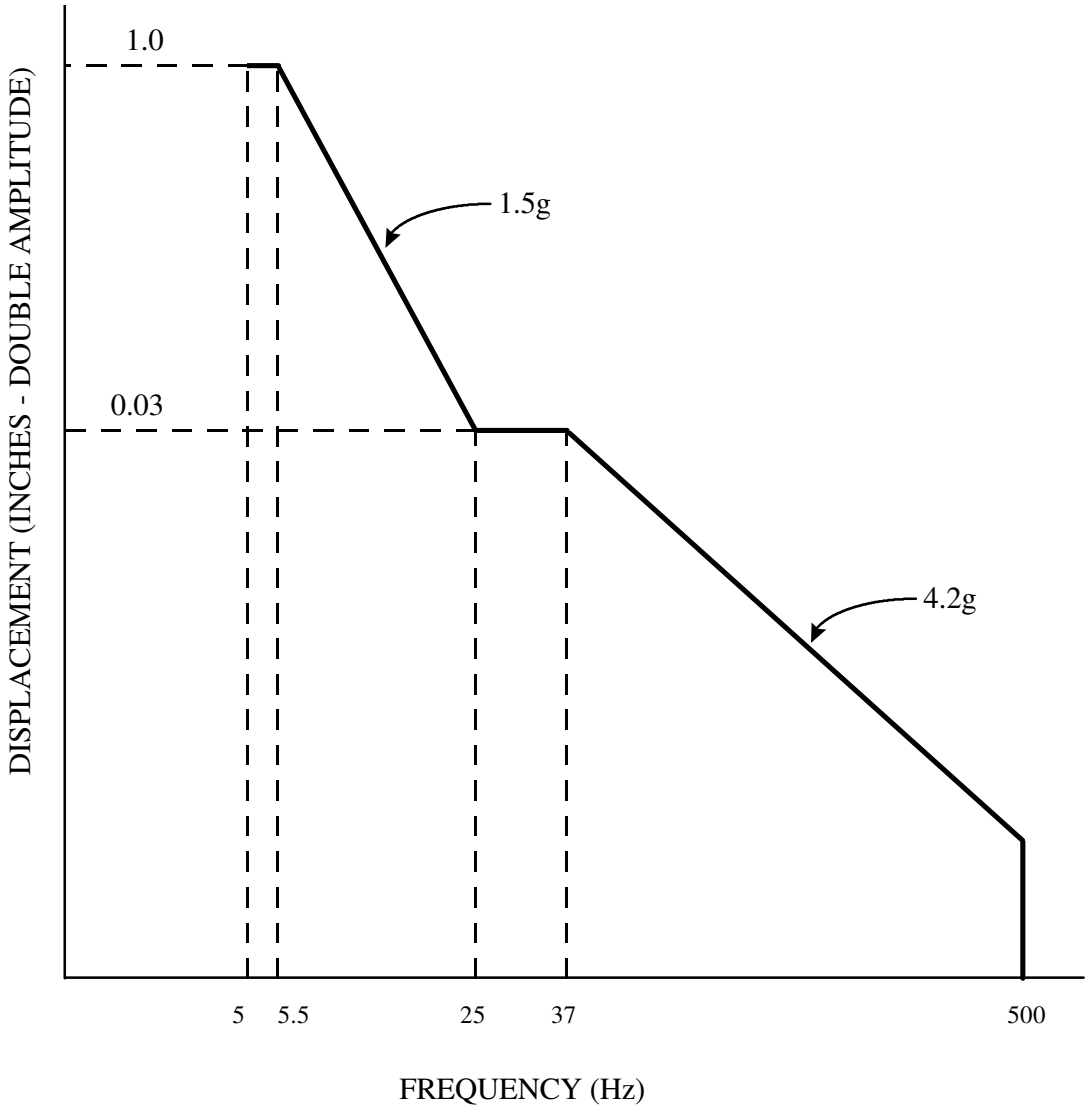
4.3.6.5 Core fin strength. To determine conformance to 3.6.5, the closely adjacent nozzle test and the distant nozzle test shall be performed as follows (see 6.9):

a. Closely adjacent nozzle test: With the point of stream contact on the radiator 1 in. (25 mm) from the nozzle outlet, and with 50 psig (345 kPa) applied to the nozzle, the stream shall strike the core surface so measured that the stream is 5° from the nominal plane of the fin.

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b. Distant nozzle test: With the point of stream contact on the radiator 36 in. (0.91 meter (m)) from the nozzle outlet and with 35 psig (241 kPa) applied to the nozzle, the stream shall strike the core in a plane normal to the core surface so measured that the stream is 45° from the nominal plane of the fin.

Permanent deformation of the fin shall not be more than 0.04 in. (1.02 mm) from its normal position shall be cause for rejection.



NOTES:
1. Acceleration levels; ± g (peak).
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 6. Vibration test curve.

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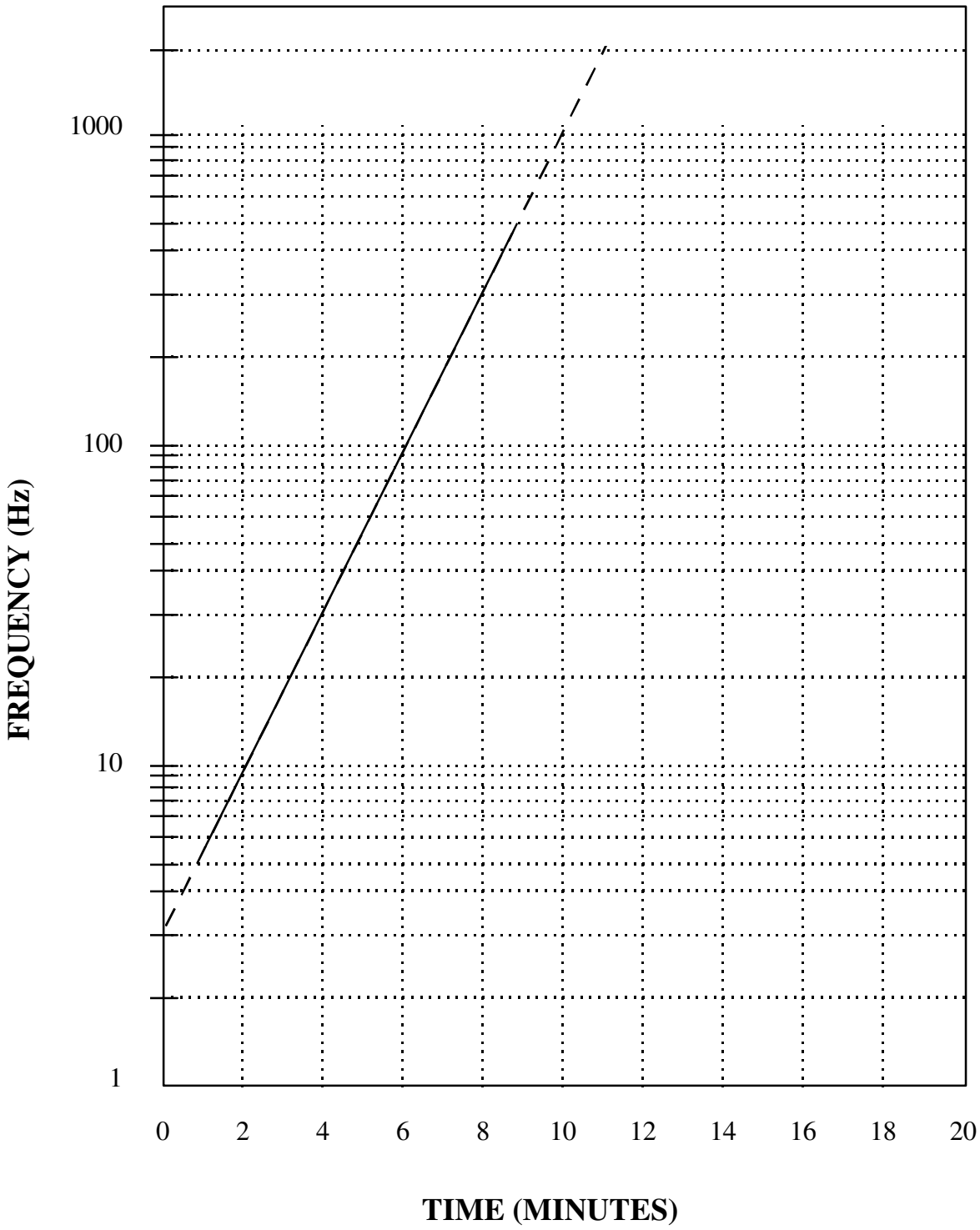


FIGURE 7. Logarithmic sweep.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains general or explanatory information that may be helpful, but is not mandatory.)

6.1 Intended use. The radiators covered by this specification are intended for use in the engine coolant systems of the vehicle classified (see 1.1). The radiator covered by this specification is military unique due to the vibrations it must be designed to endure. The vibrations resulting from the howitzer recoil and vehicle operation in battlefield terrain are more severe than a commercially available radiator would be able to successfully sustain.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification and any amendment.
b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of documents referenced (see 2.2 and 2.3).
c. When first article is required (see 3.1, 4.1.1 and 6.4).
d. Any sampling for inspection to be utilized (see 4.1.2).
e. What conformance verification is necessary (see 4.1.2 and 6.5).
f. Packaging requirements (see 5.1).
g. Any deviation from this specification.

6.3 Supersession data. When reviewing this document for changes, the previous revision "A" should not be used, as revision "A", dated 17 March 1989 was erroneously issued covering the radiator specification for the M2 and M3 vehicles. Any comparison of this document should be made against the base specification, MIL-R-62259(AT), dated 28 September 1980.

6.4 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a production item, or the number of items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article

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test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of these processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should include those tests desired from table I and their designated frequency based on a risk assessment for the procurement.

6.6 Radiator mounting. Historically, radiator has been constructed in reference to Drawing 12260209, on which figure 2 is based. Additionally, the radiator has been mounted using two each mount Drawing 11594172 and four each mount Drawing 10917099.

6.7 Lifting eye test. In the past, the hook used in this test has been similar to National Aerospace Standards NAS 1052-4.

6.8 Core fin strength. In the past, Drawing 10929606 has specified functional test equipment used for this test.

6.9 Subject term (key word) listing.

Air flow
Antifreeze
Fluid pressure
Heat transfer

6.10 Changes from previously issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - AT
Navy - MC

Review Activity:
DLA - CC

Preparing Activity:
Army - AT

(Project 2930-0031)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
	MIL-PRF-62259B	980626

3. DOCUMENT TITLE
RADIATOR: ENGINE COOLANT

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*
	(1) Commercial (810) 574-8745 (2) AUTOVON 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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